

COMSATS University Islamabad, Lahore Campus
Department of Computer Science

Assignment 3 - Agent Based Modeling (AIC332)

**Real-World Dataset Modeling in NetLogo:
Simple Model, Extended Model and Comparison**

**Topic: Agent-Based Modeling of COVID-19 Spread in Bangladesh
Using a Real-World Kaggle Dataset**

Field	Details
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Course	AIC332 - Agent Based Modeling
Tool Used	NetLogo
Dataset	COVID-19 Bangladesh dataset from Kaggle
Submission	Assignment 3 Report

1. Introduction

Agent-Based Modeling (ABM) is a simulation method in which individual agents follow simple rules and interact with each other. These small local interactions can produce large population-level patterns. This assignment uses ABM to model the spread of COVID-19 in Bangladesh using NetLogo and a real-world dataset.

The assignment required four main tasks: selecting and understanding a real-world dataset, building a simple NetLogo ABM, extending the simple model by adding at least two realistic parameters, and comparing both model outputs with the real data.

In this report, each turtle represents a group of people. A scale factor of 10 is used, so 1 turtle represents approximately 10 real people. One NetLogo tick represents one day of the dataset. The final comparison is performed against the dataset value for Day 59: 10,929 total confirmed cases and 183 total deaths.

Assignment workflow



Real data is used to design, run, and evaluate the model

Figure 1: General workflow followed in the assignment.

2. Dataset Selection and Understanding

The selected dataset is a COVID-19 Bangladesh dataset downloaded from Kaggle. It contains daily COVID-19 records from 8 March 2020 to 5 May 2020. The dataset includes confirmed cases, deaths, recovered cases, tests, and active cases.

Dataset Column	Meaning in this assignment
Day	Simulation day. One day is represented by one NetLogo tick.
Date	Calendar date of the COVID-19 observation.
Daily new confirmed cases	New infected cases reported on that day.
Total confirmed cases	Main real value used to compare model output.
Total deaths	Real death value used for extended model comparison.
Total recovered	Real recovery value used to understand the disease outcome.
Active Cases	Current infected population according to dataset.

The dataset is suitable for ABM because COVID-19 spreads through contact between individuals. Therefore, it can be represented naturally using moving agents in a NetLogo world.

3. Sample Dataset Values

The following table shows selected rows from the cleaned dataset used in the assignment. These values help show the growth trend of COVID-19 cases over time.

Day	Date	Total Cases	Total Deaths	Recovered	Active Cases
1	2020-03-08	3	0	0	3

10	2020-03-17	10	0	3	7
20	2020-03-27	48	5	11	32
30	2020-04-06	123	12	33	78
40	2020-04-16	1572	60	49	1463
50	2020-04-26	5416	145	122	5149
59	2020-05-05	10929	183	1403	9343

At the end of the dataset, Day 59 contains 10,929 total confirmed cases, 183 total deaths, 1,403 total recovered cases, and 9,343 active cases. These final values are used for the main comparison with the NetLogo models.

4. Simple ABM Design

The simple model captures only the basic disease process. Agents move randomly. If an infected agent comes near a healthy agent, the healthy agent may become infected according to the infection chance. After a fixed recovery time, infected agents become recovered.

Simple disease model



Contact with infected person -> disease spread -> recovery

Figure 2: State transition of the simple COVID-19 ABM.

Agent State	Color in NetLogo	Meaning
Healthy	Green	Person is not infected.
Infected	Red	Person has disease and can infect nearby healthy people.
Recovered	Blue	Person recovered after the recovery time.
Simple Model Parameter	Value Used	Purpose
population	1100	Total number of turtles/agents.
initial-infected	1	Starting infected turtle.
infection-chance	8	Probability of disease transmission.
recovery-time	14	Number of ticks before recovery.
scale-factor	10	1 turtle represents 10 real people.

5. Simple Model NetLogo Logic and Code Snippets

The simple model uses three main procedures: movement, infection, and recovery. The core logic is shown below.

```

to move-person
  right random 30
  left random 30
  forward 1
end

to infect-others
  ask other turtles in-radius 1 with [status = "healthy"] [
    if random-float 100 < infection-chance [
      become-infected
    ]
  ]
end
  
```

```

]
end

to recover-person
  set infected-time infected-time + 1
  if infected-time >= recovery-time [
    set status "recovered"
    set color blue
  ]
end

```

The model also loads the real dataset and compares simulated total cases with real total confirmed cases. The model error is calculated as percentage difference between simulated cases and real cases.

```

set simulated-total-cases total-infected * scale-factor
set model-error abs(simulated-total-cases - real-current-total-cases)
                / real-current-total-cases * 100

```

6. Simple Model Results

The simple model was run for 59-60 ticks, matching the number of days in the dataset. The final values are shown in the screenshot and table below.

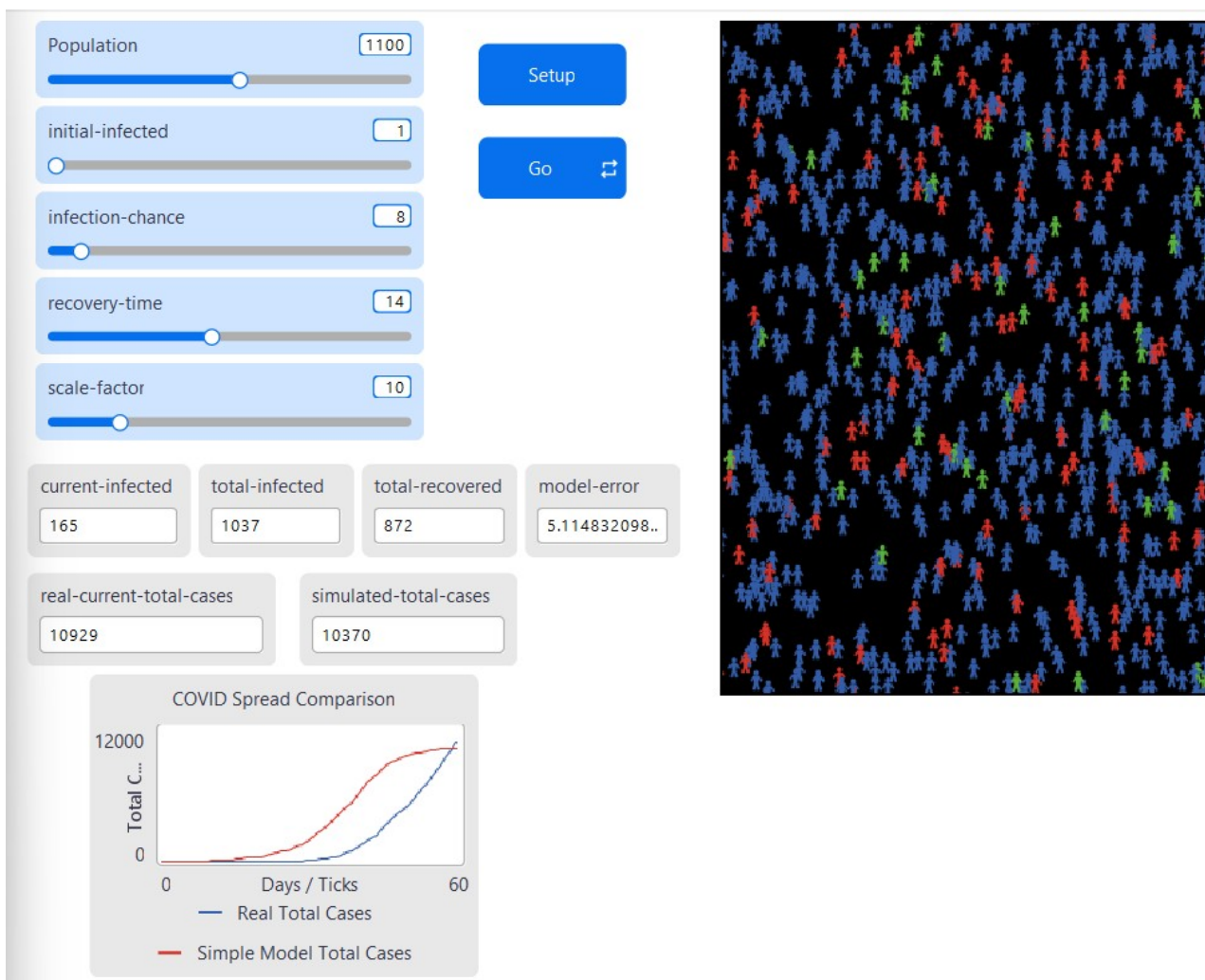


Figure 3: Simple ABM result in NetLogo showing real vs simulated total cases.

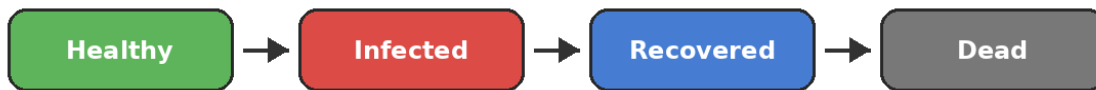
Output	Value
Real total confirmed cases	10,929
Simple model simulated cases	10,370
Current infected turtles	165
Total infected turtles	1037
Total recovered turtles	872
Case error	5.11%

The simple model produced 10,370 simulated total cases compared with 10,929 real total confirmed cases. The case error is approximately 5.11%. This shows that the simple model follows the overall growth trend of the dataset reasonably well. However, it is still a basic model because it does not include deaths, lockdown, policy changes, testing rate, age groups, or social behavior differences.

7. Extended ABM Design

The extended model improves the simple model by adding two realistic mechanisms: death chance and movement reduction. These two mechanisms make the model more realistic and satisfy the assignment requirement of adding at least two additional parameters.

Extended disease model



Adds death chance and movement reduction for more realistic behavior

Figure 4: State transition of the extended COVID-19 ABM.

Added Parameter	Value Used	Reason for Adding
death-chance	4	Real COVID-19 data includes total deaths, so infected agents may die.
movement-reduction	20	Represents lockdown/social distancing by reducing movement and contact.

In the extended model, infected agents do not always recover. After the recovery period, each infected agent either recovers or dies depending on death chance. Also, movement is reduced using the movement-reduction slider, which represents lockdown or reduced social contact.

8. Extended Model NetLogo Logic and Code Snippets

The movement reduction mechanism lowers the distance moved by each agent. For example, when movement-reduction is 20, each agent moves only 80% of the normal step size.

```

to move-person
  right random 30
  left random 30
  let step-size 1 * (1 - movement-reduction / 100)
  forward step-size

```

```
end
```

The death mechanism is added inside the disease finishing procedure. After the recovery time, an infected turtle may die or recover.

```
to finish-disease
  set infected-time infected-time + 1
  if infected-time >= recovery-time [
    ifelse random-float 100 < death-chance [
      become-dead
    ] [
      become-recovered
    ]
  ]
end

to become-dead
  set status "dead"
  set color gray
end
```

The extended model output is scaled using the same scale factor as the simple model, so simulated deaths are calculated as total-dead turtles multiplied by 10.

```
set simulated-total-cases total-infected * scale-factor
set simulated-total-deaths total-dead * scale-factor
```

9. Extended Model Results

The extended model was run using infection-chance = 12, death-chance = 4, movement-reduction = 20, recovery-time = 14, population = 1100, and scale-factor = 10.

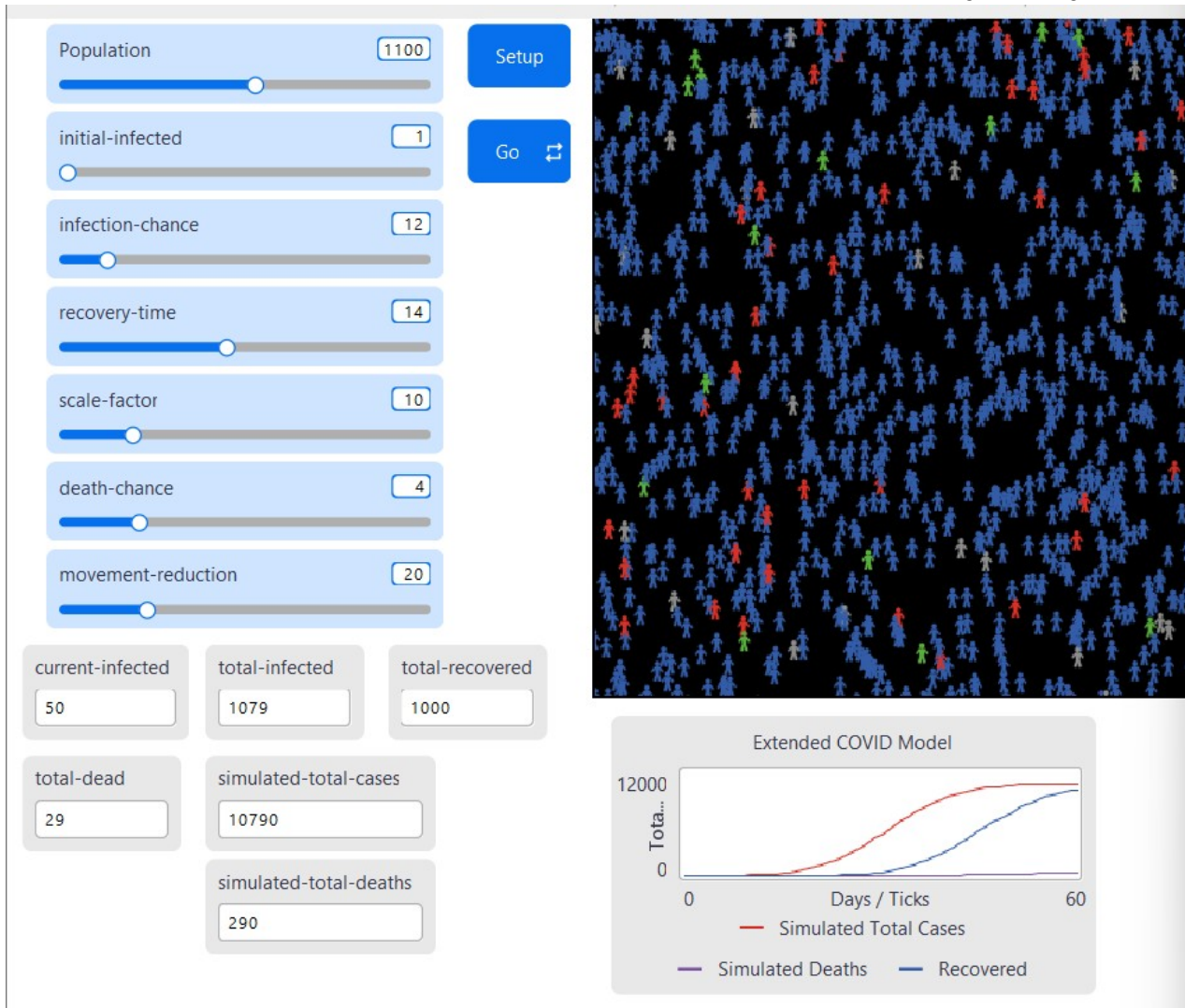


Figure 5: Extended ABM result in NetLogo showing simulated cases, deaths, and recovered people.

Output	Value
Extended model simulated cases	10,790
Extended model simulated deaths	290
Current infected turtles	50
Total infected turtles	1079
Total recovered turtles	1000
Total dead turtles	29
Case error against real data	1.27%
Death error against real data	58.47%

The extended model produced 10,790 simulated total cases, which is very close to the real value of 10,929. The case error is only 1.27%. However, the extended model produced 290 simulated deaths, while the real dataset contains 183 deaths. This means the death-chance parameter overestimated deaths in this run. This is useful because it shows that ABM results depend strongly on parameter tuning.

10. Comparison and Analysis

Output	Real Dataset	Simple ABM	Extended ABM
Total confirmed cases	10,929	10,370	10,790
Total deaths	183	Not included	290

Case error	0%	5.11%	1.27%
Death error	0%	Not applicable	58.47%
Movement control	Real world	No	Yes
Death mechanism	Real world	No	Yes
Realism level	Actual data	Basic	More realistic

The simple model is easy to understand and gives a reasonable approximation of the total cases. It shows that even simple rules like random movement, infection on contact, and recovery can produce epidemic growth patterns.

The extended model is more realistic because it adds death and movement reduction. It produced a closer total case value than the simple model. However, its death output was higher than the real dataset. This means the death-chance slider should be carefully calibrated if the goal is to match deaths accurately.

Overall, the extended model is conceptually better because it contains more real-world mechanisms. The simple model is numerically close but less realistic. This comparison shows the main idea of ABM: model behavior depends on individual rules and parameters.

11. Limitations and Possible Improvements

The model is useful for learning, but it is simplified compared with real COVID-19 spread. Real disease spread is affected by many factors that are not fully included in this model.

- Agents move randomly, while real people follow work, family, school, and travel routines.
- The model does not include age groups, hospitals, vaccination, quarantine, or contact tracing.
- The death chance is fixed, while real mortality changes with age, healthcare access, and testing accuracy.
- The infection chance is constant, while real infection risk changes with masks, indoor/outdoor contact, and policies.
- The model uses a scale factor where 1 turtle represents 10 people, so exact individual-level interpretation is limited.
- The dataset records reported cases, not all real infections, because testing capacity changes over time.

Possible future improvements include adding age groups, vaccination, different movement speeds, quarantine behavior, testing rate, social networks, and time-varying lockdown policies.

12. Conclusion

This assignment demonstrates how a real-world COVID-19 dataset can be modeled using NetLogo and Agent-Based Modeling. The simple model captured the basic disease spread process using healthy, infected, and recovered states. The extended model improved the simulation by adding death chance and movement reduction.

The simple model produced 10,370 cases with 5.11% error. The extended model produced 10,790 cases with 1.27% error, making it closer to the real total case value. However, the extended model overestimated deaths, showing the importance of parameter calibration.

The final result shows that ABM is useful for understanding how individual-level interactions can generate population-level patterns. It also shows that adding realistic mechanisms can improve a model, but only if the parameters are tuned carefully.

13. Viva / Long Question Preparation

Question	Answer
Why was COVID-19 suitable for ABM?	Because disease spread depends on interactions between individuals, which can be represented as agents.
What is one tick in the model?	One tick represents one day in the real dataset.

What does scale-factor mean?	It converts turtle counts into real people. In this model, 1 turtle = 10 real people.
Why is the extended model more realistic?	Because it includes death chance and movement reduction, which are closer to real epidemic conditions.
Why can repeated runs give different results?	The model uses randomness in movement and infection, so results can change in every run.
Why is calibration needed?	Parameters such as infection chance and death chance must be adjusted so the model matches real data better.

14. References

- COMSATS University Islamabad, Lahore Campus. Assignment 3 - Agent Based Modeling (AIC332): Real-World Dataset Modeling in NetLogo.
- Kaggle COVID-19 Bangladesh dataset used in this assignment. Cleaned file: cleaned_covid_bd_dataset.csv.
- NetLogo simulation environment used for building the simple and extended ABM models.
- Course lecture concepts: agents, environment, simple rules, emergence, simple model, extended model, and comparison analysis.